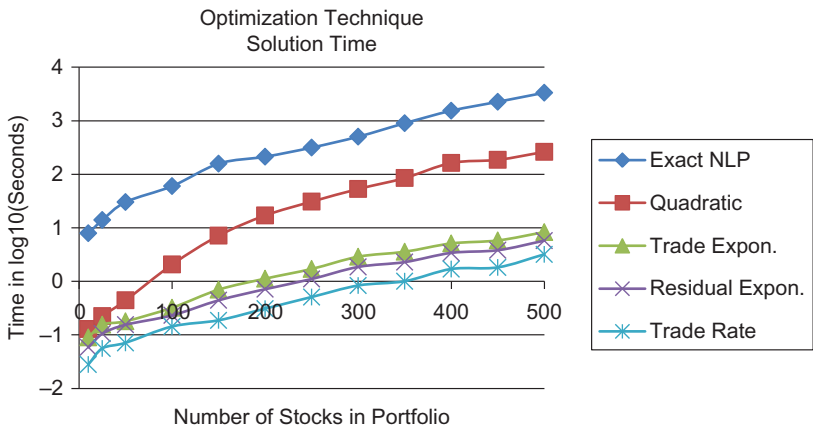


dependent upon pc, processor, and memory, analysts are encouraged to setup these experiments and analyze solution time and accuracy for the approach described above incorporating the trade list characteristics most common for their fund (e.g., small cap index, global index, growth, value, momentum, one-sided, two-sided equal, 130-30, etc).

*How long did it take to solve the portfolio objective problem?*

Figure 9.1 plots the log of the average time in seconds for each optimization routine for each portfolio size. As expected, the non-linear optimization routine, which solved for the square root risk term and a decision variable for each stock and each period, was the slowest but did provide us with the exact trade schedule to the problem. The quadratic optimization technique, which provided exact shares to trade in each period but solved for the variance of risk, was the next slowest. This technique, however, provided reasonable solution times for portfolio sizes up to about 100 stocks (analysts need to determine what is considered a reasonable solution time for their needs). A difficulty with the QP approach is that analysts need to determine the proper variance aversion parameter from the investor’s specified risk aversion parameter. So this mapping could require several runs of the problem. The trade schedule exponential and residual schedule exponential techniques provided a large improvement in solution time over the quadratic optimizer. The fastest solution was for the trade rate technique. To show the effect of the number of names in the portfolio on solution time the non-linear (NLP) optimizer took 55 minutes to solve a 500-stock portfolio. The quadratic optimizer



■ Figure 9.1 Optimization Technique Solution Time.